Yingtong Dou (Ph.D.)

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Employment

Research Scientist, Foundational AI Visa Research 12/2022 - Now
Supervisor: Dr. Mahashweta Das, Dr. Yiwei Cai Foster City, CA
• Design, train, and tune **foundation models** from scratch to serve multiple use cases in payment services; Discover and validate new use cases for foundation models
• Implement and fine-tune **Transformer**- and **LLM**-based models on multimodal transactional data for advanced payment authentication and authorization
• Develop agent-based fraud detection systems and risk modeling framework for agentic commerce and payments
• Research: foundation model for tabular/sequential data, ML trust and safety, graph mining

Research Assistant University of Illinois Chicago 06/2019 - 10/2022
Advisor: Dr. Philip S. Yu Chicago, IL
• Carried out research on graph-based fraud/anomaly/misinformation detection
• Initiated and contributed to several impactful open-source projects and libraries for graph-based outlier detection (PyGOD) and fraud detection (SafeGraph)
• Collaborated with Tencent, Grab Inc., F5 Networks, and Microsoft on various trust and safety research and application problems

Research Intern Snap Inc. 06/2021 - 10/2021
Mentor: Dr. Neil Shah Remote
• Project: probing the robustness of graph neural network under concept drift

Education

Ph.D. University of Illinois Chicago, Computer Science 2017 - 2022
Advisor: Dr. Philip S. Yu
Dissertation: *Robust Graph Learning for Misbehavior Detection* [PDF]

B.E. Beijing Univ. of Posts and Telecoms., IoT Engineering 2013 - 2017
Graduated with *Beijing Excellent Graduate Award*

Recent Work

#Citations=4,400. View complete publication list on [Google Scholar].

Y. Dou, Z. Jiang, et al. **"TransactionGPT: Toward Foundational Transaction Modeling."** *KDD*, 2026.
[doi:10.1145/3770855.3818496]

H. Le, S. Zhong, Y. Lu, **Y. Dou**. et al. **"FAFO: Lossy KV Cache Compression for Lossless Inference Acceleration via Draftless Fumble Decoding."** *ICML*, 2026.
[https://openreview.net/forum?id=kz7n1D8Y1o]

Y. Jin, K. Sharma, V. Mohan, **Y. Dou**. et al. **"SARA: Selective and Adaptive Retrieval-augmented Generation with Context Compression."** *ACL*, 2026.
[10.18653/v1/2026.acl-long.661]

K. Sharma, Y. Jin, V. Mohan, **Y. Dou**. et al. **"Sysformer: Safeguarding Frozen Large Language Models with Adaptive System Prompts."** *ICLR*, 2026.
[https://openreview.net/forum?id=EbgCEd8gyN]

Selected Work

K. Liu*, **Y. Dou***, et al. **"PyGOD: A Python Library for Graph Outlier Detection."** *JMLR*, 2024.
[JMLR:v25:23-0963]

K. Liu*, **Y. Dou***, Y. Zhao*, et al. **"BOND: Benchmarking Unsupervised Out-**

lier Node Detection on Static Attributed Graphs." *NeurIPS*, 2022.
[NeurIPS:v35:27021--27035]

Y. Dou, K. Shu, C. Xia, P. S. Yu and L. Sun. **"User Preference-aware Fake News Detection."** *ACM SIGIR*, 2021.
[doi:10.1145/3404835.3462990]

Y. Dou, Z. Liu, L. Sun, Y. Den, H. Peng, and P. S. Yu.**"Enhancing Graph Neural Network-based Fraud Detectors against Camouflaged Fraudsters."** *ACM CIKM*, 2020.
[doi:10.1145/3340531.3411903]

Y. Dou, G. Ma, P. S. Yu, and S. Xie.**"Robust Spammer Detection by Nash Reinforcement Learning."** *ACM SIGKDD*, 2020.
[doi:10.1145/3394486.3403135]

Professional Services

- **Conference Organizer:** TheWebConf'22 Whova Chair, WSDM'22 Whova Chair
- **Session Chair:** CIKM'22, KDD'22
- **Journal Reviewer:** IEEE: TPAMI, TBD, TIFS, TKDE, TNNLS; ACM: TWEB, TKDD, TIST, TOIS; Springer: MTA, IPM, SNAM; Elsevier: KBS, OSNM, IAM, PR; Frontiers in: Computer Science, Psychology
- **Program Committee Member:** KDD, NeurIPS, SIGIR, IJCAI, TheWebConf, AAAI, ACL, EMNLP, CIKM, WSDM, RecSys, ICWSM, ICDM, ECML-PKDD, SDM, LoG, ASONAM

Talks and Blogs

- **GNN-based Fraud Detection: from Research to Application.**
  Wells Fargo, Online. [*Slides PDF*] December 2022
- **GNN-based Graph Outlier Detection: A Brief Introduction.**
  Blog Article, TigerGraph. [*Blog Link*] September 2022
- **GNN-based Anomaly Detection: from Research to Application.**
  One Novartis Data Science Seminar, Online. [*Slides PDF*] August 2022
- **GNNs for Financial Fraud Detection: Practices and Challenges.**
  KDD'22 MLF Workshop, Washington DC. [*Slides PDF*] August 2022
- **Mining Twitter for Social Event and Misinformation Detection.**
  CS 579 OSN Analysis, IIT, Chicago. [*Slides PDF*] March 2022
- **Robust Graph Learning for Misbehavior Detection.**
  Ph.D. Forum, ACM WSDM 2022, Online. [*Slides PDF*] February 2022
- **GNN-based Fraud Detection: from Research to Application.**
  Datafun Summit, Online. [*Slides PDF*] October 2021
- **Robust Fraud Detection against Adversarial Fraudsters.**
  Grab Inc. and F5 Research, Online. [*Slides PDF*] March & May 2021
- **Tackling Fake Downloads in Mobile App Markets.**
  Blog Article, Medium. [*Blog Link*] July 2019
- **An Introduction to Graph-based Spam Review Detection.**
  Tencent, Shenzhen. [*Slides PDF*] July 2018

Honors and Awards

- ACM CIKM 2022 Best Paper Honorable Mention October 2022
- Papers with Code Contributor Award March 2022
- ACM WSDM 2022 Outstanding Service Award February 2022
- Student Travel Awards: WSDM'22, ICDM'21, SIGIR'20, SIGKDD'20, CIKM'20